

RED DEER, AB, Canada

WMO: 718780

Lat: 52.179N

Lon: 113.893W

Elev: 905

StdP: 90.92

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-27.7	-36.1	0.1	-31.2	-32.6	0.2	-27.6	10.4	-6.5	9.1	-8.5	2.0	200	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	27.9	17.2	26.1	16.4	24.4	15.6	18.8	25.6	17.6	24.2	16.5	22.7	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.8	22.3	15.0	11.9	21.0	13.9	11.0	19.7	57.1	25.7	53.2	24.3	49.6	22.7	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
9.2	8.2	7.2		-35.8	31.4	3.8	1.8	-38.6	32.7	-40.8	33.8	-43.0	34.8	-45.8	36.1	
				-36.0	20.6	3.8	1.4	-38.7	21.6	-40.9	22.4	-43.0	23.2	-45.8	24.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.4	-10.0	-4.4	3.5	9.7	13.7	16.4	15.2	10.5	3.8	-4.5	-10.4
	DBStd	11.59	8.38	7.47	7.74	5.10	4.37	2.97	2.89	3.18	4.19	4.68	6.93	7.60
	HDD10.0	3237	662	560	447	198	58	4	1	2	42	196	434	632
	HDD18.3	5707	921	793	705	444	267	141	75	103	234	449	683	891
	CDD10.0	595	0	0	0	4	50	116	197	164	58	4	1	0
	CDD18.3	24	0	0	0	0	1	3	13	7	1	0	0	0
	CDH23.3	602	0	0	0	2	42	61	225	203	68	2	0	0
	CDH26.7	116	0	0	0	0	4	7	47	44	12	0	0	0
Wind	WSAvg	3.1	2.8	2.7	3.1	3.7	3.8	3.4	2.8	2.6	2.9	3.2	2.9	2.8
Precipitation	PrecAvg	506	24	17	22	28	61	99	86	66	40	21	21	21
	PrecMax	611	60	43	48	87	127	190	214	116	80	51	62	49
	PrecMin	361	6	4	3	4	26	21	22	15	6	7	3	6
	PrecStd	62	12	8	11	18	23	33	38	28	23	13	14	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	6.8	13.6	22.2	27.1	27.4	30.3	30.5	28.4	22.4	14.7	7.8
		MCWB	2.7	2.4	6.5	10.5	13.5	16.6	18.8	17.8	15.5	11.4	7.6	2.7
	2%	DB	4.7	4.3	9.6	18.8	24.3	25.0	28.0	27.6	25.2	18.1	10.0	4.9
		MCWB	1.5	0.8	4.2	9.2	12.6	15.2	18.5	17.1	14.6	9.8	4.8	1.4
	5%	DB	2.4	2.3	7.1	15.7	21.7	23.1	25.8	25.6	22.6	14.9	6.7	2.4
		MCWB	-0.2	-0.6	2.8	7.2	11.6	14.3	17.6	16.5	13.5	8.0	2.7	-0.4
	10%	DB	0.0	0.3	5.0	12.7	19.2	21.1	23.9	23.5	19.7	12.1	4.4	0.2
		MCWB	-2.1	-1.9	1.5	5.7	10.3	13.4	16.7	15.8	12.3	6.6	1.3	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.8	7.0	11.0	14.6	18.1	20.8	19.6	16.7	12.3	7.8	3.2
		MCDB	6.3	6.2	13.4	21.2	24.2	25.2	27.5	26.8	25.1	20.7	14.6	6.9
	2%	WB	1.5	1.0	4.6	9.4	13.2	16.4	19.4	18.3	15.1	10.1	5.1	1.5
		MCDB	4.5	4.0	9.1	18.1	22.4	22.8	26.1	25.3	23.5	17.4	9.5	4.6
	5%	WB	0.0	-0.4	3.0	7.7	12.0	15.2	18.2	17.2	13.9	8.5	3.0	-0.2
		MCDB	2.3	2.1	6.9	14.9	20.8	21.5	24.8	24.1	21.9	14.2	6.2	2.3
	10%	WB	-2.1	-1.9	1.5	6.0	10.8	14.1	17.1	16.1	12.6	6.9	1.4	-1.9
		MCDB	0.0	0.3	4.7	11.9	18.2	20.0	22.9	22.9	19.4	11.5	4.1	0.2
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.5	10.6	12.1	13.0	11.5	12.6	13.4	13.7	12.7	10.5	10.4
		MCDBR	12.5	13.3	13.3	18.7	17.9	15.7	16.2	18.1	19.9	18.0	14.0	12.2
		MCWBR	9.9	10.6	9.0	10.4	8.7	7.7	8.4	9.2	10.3	10.8	9.6	9.4
	5% WB	MCDBR	12.2	13.0	13.1	17.4	16.5	13.6	15.1	16.2	18.3	17.2	13.1	12.1
		MCWBR	9.7	10.4	9.1	10.0	8.2	7.2	8.3	8.7	9.7	10.5	9.3	9.5
Clear-Sky Solar Irradiance	taub	0.261	0.267	0.293	0.328	0.338	0.345	0.346	0.359	0.309	0.282	0.253	0.244	
	taud	2.330	2.406	2.420	2.363	2.390	2.398	2.418	2.365	2.477	2.537	2.476	2.330	
	Ebn at Noon	757	870	910	909	913	907	900	867	881	846	779	715	
	Edh at Noon	60	74	90	109	111	112	108	108	85	65	51	50	
All-Sky Solar Radiation	RadAvg	1.07	2.06	3.45	4.67	5.60	5.94	6.31	5.18	3.62	2.18	1.14	0.81	
	RadStd	0.08	0.16	0.24	0.40	0.38	0.40	0.35	0.35	0.39	0.24	0.10	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page